TTA — Production Deployment SOP

tta.indiakhelofootball.com

Version	1.0
Date	February 2026
Server OS	Ubuntu 18.04.6 LTS
Stack	Django · Gunicorn · Nginx · MariaDB 10.1
Domain	tta.indiakhelofootball.com

INDEX

1. SERVER IDENTITY

This is the machine you are deploying to. Before you run a single command, confirm you are connected to this server.

What	Value
Hostname	iZt4n8bk3wgx3xb2uxq5rrZ
Operating System	Ubuntu 18.04.6 LTS (Bionic)
Kernel	4.15.0-162-generic
CPU	Intel Xeon Platinum
RAM	3.7 GB total / ~1.2 GB available
Swap	2.0 GB (enabled)
Disk	69 GB total / 45 GB used (68%) — monitor this
Database	MariaDB 10.1.48
Domain	tta.indiakhelofootball.com

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2. PROJECT LAYOUT ON THE SERVER

Everything lives inside /root/TTA/. Every command in this document uses these exact paths. Learn them.

Path	What It Is
/root/TTA/	Root folder for the entire project
/root/TTA/backend/ikf-tta-backend/backend	Django application code
/root/TTA/backend/venv	Python virtual environment — all packages live here
/root/TTA/frontend/ikf-tta-frontend	React frontend source code
/root/TTA/frontend/ikf-tta-frontend/build	Built frontend files — Nginx serves from here

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3. HOW THE APPLICATION RUNS

Traffic comes in, hits Nginx first. Nginx handles HTTPS and passes requests to Gunicorn through a socket file. Gunicorn runs the Django code. Django talks to MariaDB. Here is the exact chain:

```
User Browser → Nginx (port 443, HTTPS) → /run/tta.sock → Gunicorn (3 workers) → Django →
MariaDB
```

systemd Service File

Gunicorn is managed by systemd under the service name 'tta'. Service file location: /etc/systemd/system/tta.service

```
[Unit]
Description=gunicorn tta daemon
Requires=tta.socket
After=network.target

[Service]
User=root
Group=www-data
WorkingDirectory=/root/TTA/backend/ikf-tta-backend/backend
ExecStart=/root/TTA/backend/venv/bin/gunicorn \
--access-logfile - \
--workers 3 \
--bind unix:/run/tta.sock \
backend.wsgi:application

[Install]
WantedBy=multi-user.target
```

Nginx Site Configuration

Site file: /etc/nginx/sites-available/tta.indiakhelofootball.com SSL certificate: /etc/letsencrypt/live/indiakhelofootball.com/fullchain.pem

```
server {
listen 80;
server_name www.tta.indiakhelofootball.com tta.indiakhelofootball.com;
return 301 https://tta.indiakhelofootball.com$request_uri;
}

server {
listen 443;
server_name www.tta.indiakhelofootball.com tta.indiakhelofootball.com;
client_max_body_size 16M;
root /root/TTA/frontend/ikf-tta-frontend/build;

location / { try_files $uri $uri/ /index.html; }
location /api/ { include proxy_params; proxy_pass http://unix:/run/tta.sock; }
location /admin/ { include proxy_params; proxy_pass http://unix:/run/tta.sock; }

ssl_certificate /etc/letsencrypt/live/indiakhelofootball.com/fullchain.pem;
ssl_certificate_key /etc/letsencrypt/live/indiakhelofootball.com/privkey.pem;
ssl_protocols TLSv1 TLSv1.1 TLSv1.2;
}
```

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4. BEFORE YOU DEPLOY — CHECKLIST

Go through every item before you SSH into the server. If any item is not done, stop and fix it first.

■ Code has been reviewed by at least one other person.

■ Changes are merged into the main branch.

■ You have tested the changes and nothing is broken.

■ If you changed the database schema, the migration script is written and tested.

■ You have taken a database backup (see Section 7).

■ You know the last stable commit hash in case you need to roll back.

■ You have told the team you are about to deploy.

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5. DEPLOYMENT STEPS

Follow these steps in order. Do not skip any step.

Step 1 — SSH Into the Server

Connect to the server. Once in, confirm the hostname.

```
ssh root@<server-ip>
hostname
```

You should see: iZt4n8bk3wgx3xb2uxq5rrZ

Step 2 — Take a Database Backup

Do this before you touch any code. If something breaks, this backup is how you recover.

```
mysqldump -u root -p --single-transaction --quick tta > /root/backups/tta_backup_$(date
+%Y%m%d_%H%M%S).sql
ls -lh /root/backups/
```

Step 3 — Pull the Latest Code

```
cd /root/TTA
git pull origin main
```

If git shows merge conflicts, stop here. Resolve the conflict on your local machine, push, then come back and pull again.

Step 4 — Install Backend Dependencies

```
source /root/TTA/backend/venv/bin/activate
pip install -r /root/TTA/backend/ikf-tta-backend/backend/requirements.txt
```

Wait for it to finish. If any package fails, stop and fix it before continuing.

Step 5 — Run Database Migrations

Run this step if you changed the database schema. It is safe to run even if nothing changed.

```
cd /root/TTA/backend/ikf-tta-backend/backend
python manage.py migrate
```

Read the output. If it says 'No migrations to apply' or lists applied migrations, you are good. If it shows an error, stop. Do not restart the service. Go to Section 6 (Rollback).

Step 6 — Build the Frontend

Run this only if you made changes to the React frontend. Skip it if only backend changed.

```
cd /root/TTA/frontend/ikf-tta-frontend
npm install
npm run build
```

The build output goes into the /build folder. Nginx already points there.

Step 7 — Restart the Backend Service

```
systemctl restart tta
systemctl status tta
```

Look for 'active (running)' in the output. If you see 'failed' for more than 10 seconds, go to Section 6 (Rollback) immediately.

Step 8 — Verify Nginx Is Healthy

```
nginx -t
```

You should see: 'syntax is ok' and 'test is successful'.

Step 9 — Check the Logs

Watch the logs for 2 to 3 minutes after restarting.

```
journalctl -u tta -f
```

You are looking for normal request logs and no Python tracebacks. Press Ctrl+C to exit once satisfied.

Step 10 — Manual Smoke Test

Open a browser and test the live site:

— Open https://tta.indiakhelofootball.com and confirm the page loads.

— Log in and confirm the core features work.

— Hit /api/reps/ and /api/trials/ and confirm they return data.

If everything is working, deployment is done. Tell the team.

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6. HOW TO ROLL BACK

Something broke. Stay calm. Do not make random changes. Follow these steps in order.

Step 1 — Find the Last Working Commit

```
cd /root/TTA
git log --oneline -10
```

This shows the last 10 commits. Find the one that was working before your deployment. Copy its short hash.

Step 2 — Revert the Code

```
git checkout <commit-hash>
```

Step 3 — Restore the Database If Migrations Were Run

Only do this if you ran migrations during the failed deployment. If no migrations ran, skip this step.

```
mysql -u root -p tta < /root/backups/tta_backup_YYYYMMDD_HHMMSS.sql
```

Use the filename of the backup you created before this deployment.

Step 4 — Reinstall Dependencies and Restart

```
source /root/TTA/backend/venv/bin/activate
pip install -r /root/TTA/backend/ikf-tta-backend/backend/requirements.txt
systemctl restart tta
systemctl status tta
journalctl -u tta -f
```

Confirm the application is running and the site is working again. Tell the team what happened.

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7. DATABASE BACKUPS

Take a backup before every deployment. A backup you have not tested is not a backup. Keep backups off the server when possible.

Manual Backup

```
mysqldump -u root -p --single-transaction --quick tta > /root/backups/tta_backup_$(date
+%Y%m%d_%H%M%S).sql
ls -lh /root/backups/
```

Automated Daily Backup via Cron

```
crontab -e
```

Add this line to run a backup every night at 2 AM:

```
0 2 * * * mysqldump -u root -p<password> --single-transaction tta >
/root/backups/tta_backup_$(date +\%Y\%m\%d).sql
```

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8. HEALTH CHECKS

Run these after every deployment and at least once a week on their own.

Service Status

```
systemctl status tta
systemctl status nginx
systemctl status mariadb
```

Disk Space

```
df -h
```

If /dev/vda1 hits 80% or more, clean up old backups or logs immediately. Currently at 68%.

Memory

```
free -h
```

If available memory drops below 200 MB, the server will start using swap and slow down.

Application Logs

```
journalctl -u tta -n 100
```

Look for Python errors, database connection failures, or unusual 500 responses.

SSL Certificate

```
certbot renew --dry-run
```

Run this monthly. Certificates expire every 90 days. Verify auto-renewal is working.

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9. WHEN SOMETHING GOES WRONG — INCIDENT LEVELS

Level	What It Means	What To Do
P0 — Critical	Site is completely down. No one can access it.	Roll back immediately. Tell everyone. Fix within 1 hour.
P1 — Major	Core feature broken (login, saving data, etc.).	Roll back or hotfix. Inform team lead. Fix same day.
P2 — Minor	Something looks wrong but core features work.	Log it. Fix in next deployment. No rollback needed.

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10. KNOWN ISSUES ON THIS SERVER

These are real issues that exist on this server right now. They are documented here so the team is aware of them.

— Gunicorn runs as root

The service file sets User=root. Create a dedicated deploy user and update the service file.

— Project lives in /root/TTA

Applications should not run from the root home directory. Move to /srv/tta/ and update paths in the service file and Nginx config.

— Ubuntu 18.04 is end of life

No security patches are being released for this version. Plan an upgrade to Ubuntu 22.04 LTS.

— MariaDB 10.1 is no longer supported

This version is from 2016 and has known vulnerabilities. Upgrade to MariaDB 10.6 LTS or 10.11 LTS.

— TLS 1.0 and 1.1 are enabled

These are deprecated. In the Nginx config, change ssl_protocols to: TLSv1.2 TLSv1.3

— Service is marked as disabled

If the server reboots, the service will not start automatically. Run: systemctl enable tta

— Socket has world-writable permissions (777)

Any process on the server can write to /run/tta.sock. Add UMask=007 to the service file.

— No restart policy in service file

If Gunicorn crashes, systemd will not restart it. Add Restart=always and RestartSec=5 to the [Service] section.

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11. QUICK REFERENCE — COMMANDS

What You Want To Do	Command
Restart backend	systemctl restart tta
Check backend status	systemctl status tta
Read backend logs (live)	journalctl -u tta -f
Read last 100 log lines	journalctl -u tta -n 100
Test Nginx config	nginx -t
Reload Nginx	systemctl reload nginx
Restart Nginx	systemctl restart nginx
Restart database	systemctl restart mariadb
Check disk space	df -h
Check memory	free -h
Activate virtual environment	source /root/TTA/backend/venv/bin/activate
Go to backend code	cd /root/TTA/backend/ikf-tta-backend/backend
Go to frontend code	cd /root/TTA/frontend/ikf-tta-frontend
Run DB migrations	python manage.py migrate
Build frontend	npm run build
Take DB backup	mysqldump -u root -p --single-transaction tta > /root/backups/backup.sql
Enable service on boot	systemctl enable tta
Reload systemd after edit	systemctl daemon-reload
View last git commits	git log --oneline -10
Revert to a commit	git checkout <commit-hash>

Update this document whenever you change anything significant on the server — service file, Nginx config, project paths, or database name. An outdated SOP is worse than no SOP.